MILITARY SPECIFICATION SHEET

ELECTRON TUBES, RECEIVING

TYPES 12SN7GT AND 12SX7GT 1/

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Twin triode, medium Mu

Outline	---	9-11 or 9-41 (EIA)
Base	---	B8-6, B8-46, or B8-58
Envelope	---	T9
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	2g	2a	2k	1g	1a	1k	h	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ik/k	Pp/a	Ⓑ Ehk	Ⓑ Alt
Unit:	V	Vdc	Vdc	mAdc	W	v	ft
Maximum:	13.8	330	---	20	2.75	100	(See note 5)
Minimum:	11.4	28	---	---	---	---	
TEST CONDITIONS:	12.6	250	-8	---	---	---	---

GENERAL:

Qualification - Required

1/ See note 2

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Quality conformance inspection, part 1					
1266	Total grid current	See notes 1 and 3	Ic	0	-2.0	µAdc
1306	Transconductance (1)	See note 4	Sm	2,075	3,125	µmhos
1306	Transconductance (3) Type 12SX7GT	Eb = 28 Vdc; Ec = 0; Rg = 0.05 Meg (see notes 4 and 6)	Sm	1,400	2,200	µmhos
1246	Audio frequency noise	Esig = 70 mVac; Rp = 10,000 ohms (see note 1)	EB	---	17	vu
1231	Emission	Eb = Ec = 30 Vdc (see notes 3 and 4)	Is	40	---	mAdc
(B) 1201	Short and discontinuity detection		---	---	---	---
1256	Electrode current (1) (anode)	See note 4	Ib	5.5	12.5	mAdc
	Quality conformance inspection, part 2					
1031	Low-frequency vibration	Rp = 2,000 ohms (see note 1)	Ep	---	150	mVac
1301	Heater current		If	275	325	mA
1336	Heater-cathode leakage	See note 1	Ihk	0	30	µAdc
1306	Transconductance (2)	Eb = 90 Vdc; Ec = 0 (see note 4)	Sm	2,400	3,600	µmhos
1256	Electrode current (2) (anode)	Ec = -24 Vdc (see note 4)	Ib	0	20	µAdc
1316	Amplification factor	See note 4	Mu	18	23	---
(B) 1101	Secureness of base, cap, or insert		---	---	---	---
(B) 1105	Permanence of marking		---	---	---	---
	Quality conformance inspection, part 3					
---	Life-test provisions	Group A; Ehk = 100 V (see note 1)	---	---	---	---
---	Life-test end point (500 hours)	Transconductance (1)	Sm	1,550	---	µmhos

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NOTES:

1. Test units connected in parallel.
2. (B) Tube type 6SN7WGT is hereby deleted from this specification sheet. For replacement purposes use tube type 6SN7WGTA, MIL-E-1/663.
3. This test to be performed at the conclusion of the holding period.
4. Test each unit separately.
5. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
6. Grid resistor shall be properly by-passed.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MU
Navy -
Air Force - 11, 85
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2401-74)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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